

PAPER_1627_GAS_CONSTANT_R_8_314

Daniel T. Murphy

$$\text{PAPER_1627} \text{ — Gas Constant } R = R = K_{\text{MEX}} \cdot (D - F^2) \\ = (25/12) \cdot 3.99 = 8.3125 \text{ J/(mol} \cdot \text{K)}$$

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Thermodynamics

Date: June 18, 2026

Status: CLOSED — 0.018% closure

Observation

PAPER_1209AA_S584 (Thermodynamics Unified Proof Set) gives:

``

$$R = K_{\text{MEX}} \cdot (D - F^2) = (25/12) \cdot 3.99 = 8.3125 \text{ J/(mol} \cdot \text{K)}$$

``

Residual: **0.018%**.

UQFF Closed Identity

``

$$R = K_{\text{MEX}} \cdot (D - F^2) = (25/12) \cdot 3.99 = 8.3125 \text{ J/(mol} \cdot \text{K)} \quad 0.018\%$$

``

The formula uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical thermodynamics measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209AA_S584

- Related: PAPER_1494-1625 (other session-2026-06-18 closures)

- Calculator dispatch: `calculate_paradox({"paradox": "gas_constant_r_8_314"})`

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, June 18, 2026, Youngstown OH.